

MEDTRACK_DV

Hospital Operations & Patient Analytics Dashboard

Project Documentation

Infosys Springboard 7.0 – Data Visualization (Batch -2)

Implementation using Power BI and Tableau

Prepared by: Pricilla G

Arunachala College of Engineering for Women

B.Tech – Artificial Intelligence & Data Science

GitHub Branch: Pricilla-G

1. Project Overview

MedTrack_DV is a hospital operations and patient analytics project developed as part of Infosys Springboard 7.0. The project takes hospital data through data preparation, KPI generation and dashboard development to present useful information through Power BI and Tableau.

The project is organized into eight modules so that each stage has a clear input and output.

Module	Work
Module 1	Data Collection / Generation and Integration
Module 2	Data Cleaning and Preparation
Module 3	Hospital KPI Engineering
Module 4	Dashboard Planning and Storyboard
Module 5	Hospital Overview and Patient Flow
Module 6	Department Analytics and Resource Utilization
Module 7	Final Integration, Testing and Validation
Module 8	Documentation and Project Delivery

Four Dashboard Areas

Dashboard	Main purpose
Hospital Overview	Overall hospital performance and major KPIs
Patient Flow	Admissions, patient stay and readmission analysis
Department Analytics	Department-level volume, performance and efficiency
Resource Utilization	Bed, ward and patient-load analysis

2. Problem Statement and Objectives

Hospitals generate information from patient admissions, department activity, bed status and other operational activities. When these measures are viewed separately, it can be difficult to understand the overall operational situation.

MedTrack_DV brings the available measures into an analytical environment so that hospital information can be viewed through KPIs, charts, filters and dashboard interactions.

Objectives

- Prepare a structured hospital dataset for analysis.
- Clean and standardize the available data.
- Calculate important hospital KPIs.
- Plan the dashboard layout before development.
- Develop four dashboards covering major hospital operations.
- Use Power BI and Tableau for dashboard implementation.
- Test calculations, filters and dashboard behavior.
- Organize and document the final project for GitHub delivery.

3. Tools and Technologies

Tool / Technology	Use in the project
Python	Data preparation and KPI-related processing
Pandas / NumPy	Data cleaning, transformation and numerical processing
Jupyter Notebook	Hospital data-cleaning workflow
CSV	Hospital raw/cleaned datasets
Excel	Final analytical dataset and KPI summaries
Power BI	Dashboard development
Tableau	Dashboard development
GitHub	Project organization and delivery
PDF	Dashboard storyboard and project documentation

4. Module 1 – Data Collection / Generation & Integration

Module 1 establishes the data foundation for the project. The hospital information is collected or prepared and arranged into a raw dataset for the cleaning stage.

- Prepare the required hospital and patient admission information.
- Collect or generate the required data fields.
- Integrate the information into the raw hospital dataset.
- Use the raw dataset as the input for Module 2.

File	Purpose
data_collection.py	Python script used for the data collection / preparation stage.
hospital_raw_data.csv	Raw hospital dataset used as the source for cleaning.

Module 2 – Data Cleaning & Preparation

The cleaning stage converts the raw data into a consistent dataset suitable for KPI generation and dashboard analysis. The submitted Jupyter Notebook contains the cleaning workflow.

- Check missing and duplicate values.
- Standardize column names and categorical values.
- Standardize department and gender information.
- Convert and validate date fields.
- Validate patient age and create useful age-related fields.
- Prepare time-related fields for later analysis.
- Save the cleaned dataset as a separate CSV file.

File	Purpose
hospital_cleaning.ipynb	Notebook used for data cleaning and preparation.
hospital_cleaned.csv	Cleaned dataset used by the later modules.

5. Module 3 – Hospital KPI Engineering

Module 3 uses the cleaned dataset to calculate the hospital KPIs needed for dashboard analysis. The KPI-generation script also creates additional analytical fields and summary sheets.

File	Purpose
generate_hospital_kpis.py	Python script used for hospital KPI generation.
hospital_final_dataset.xlsx	Final analytical workbook containing the dataset and KPI summaries.

KPIs generated

KPI	Meaning
Total Admissions	Total number of admission records.
Occupancy Rate	Hospital occupancy indicator.
Average Length of Stay	Average duration of patient stay.
Readmission Rate	Percentage of patients identified as readmitted.
Bed Utilization Rate	Bed-use indicator based on the prepared data.
Department Efficiency Score	Relative department performance score.

KPI workbook sheets

- Final_Dataset
- KPI_Summary
- KPI_Definitions
- Department_Analytics
- Monthly_KPIs

6. Module 4 – Dashboard Planning & Storyboard

Module 4 is the planning stage for the dashboard development. The storyboard is used to decide the structure, information placement and analytical flow before final dashboard work.

Dashboard	Planned focus
Hospital Overview	Hospital KPIs, admissions and overall performance
Patient Flow	Admissions, stay-related measures and readmission
Department Analytics	Department volume, LOS and efficiency comparison
Resource Utilization	Beds, ward types, patient load and resource analysis

Deliverable: dashboard_Storyboard.pdf

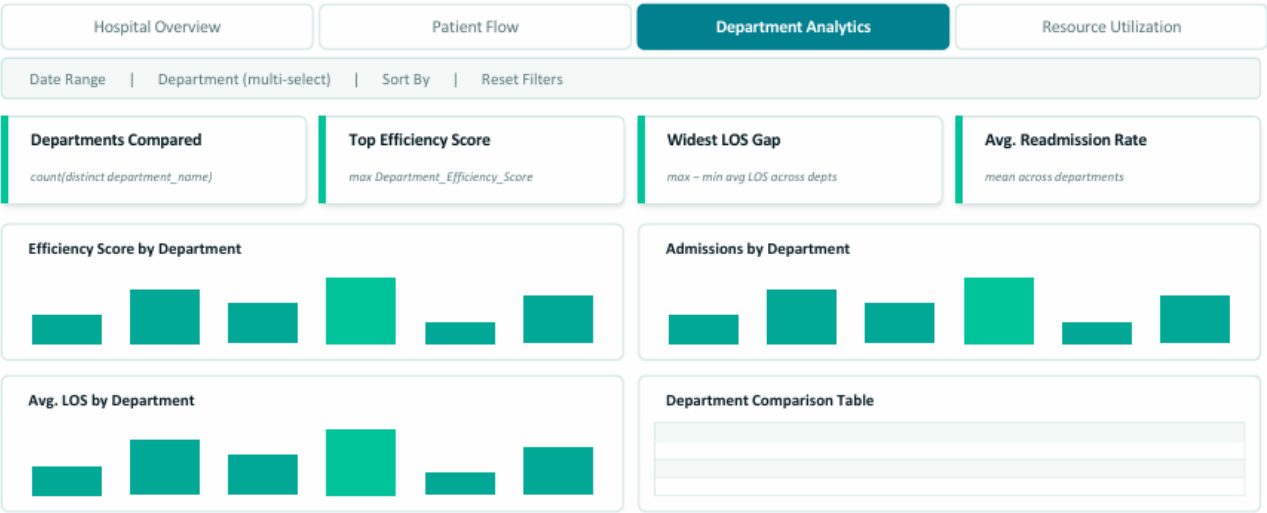
Hospital Overview



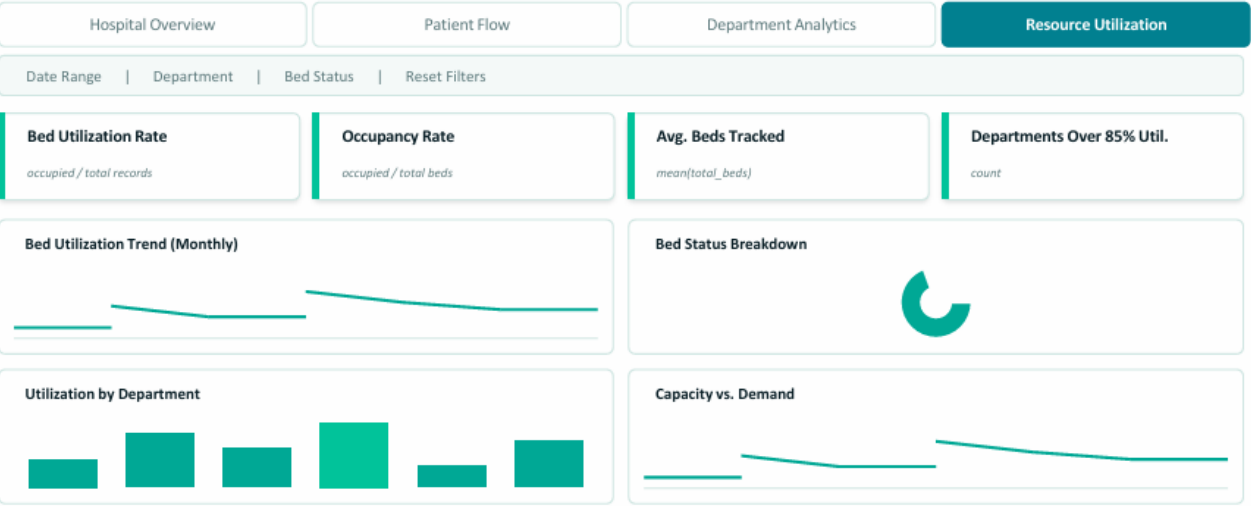
Patient Flow



Department Analytics



Resource Utilization



7. Module 5 – Hospital Overview & Patient Flow

Module 5 covers the first two dashboard areas. The dashboard development version in the repository is `medtrack_dashboard_v1.pbix`.

Hospital Overview

- Main hospital KPIs.
- Admission type mix.
- Patient distribution by department.
- Admission analysis by year.
- Disease-category analysis.
- Hospital-level operational measures.

Patient Flow

- Admission trends by month.
- Length-of-stay distribution.
- Patient stay distribution.
- Readmission-status analysis.

8. Module 6 – Department Analytics & Resource Utilization

Module 6 covers the remaining two dashboard areas and completes the planned four-dashboard scope.

Department Analytics

- Average length of stay by department.
- Admission count by department.
- Department performance comparison.
- Admission by gender.
- Readmission trend by year.

Resource Utilization

- Department contribution to hospital load.
- Patient load compared with length of stay.
- Length of stay by admission type.
- Bed status by ward type.
- Patient load by ward type.

Correction for folder organization: do not duplicate the same final PBIX/TWBX file in Modules 5 and 6. Keep `medtrack_dashboard_v1.pbix` as the development version under Module 5, and keep the final `MedTrack_DV.pbix` and `MedTrack_DV.twbx` under Module 7. If separate Module 6 prototype files exist, they should be placed in Module 6 instead.

9. Module 7 – Final Integration, Testing & Validation

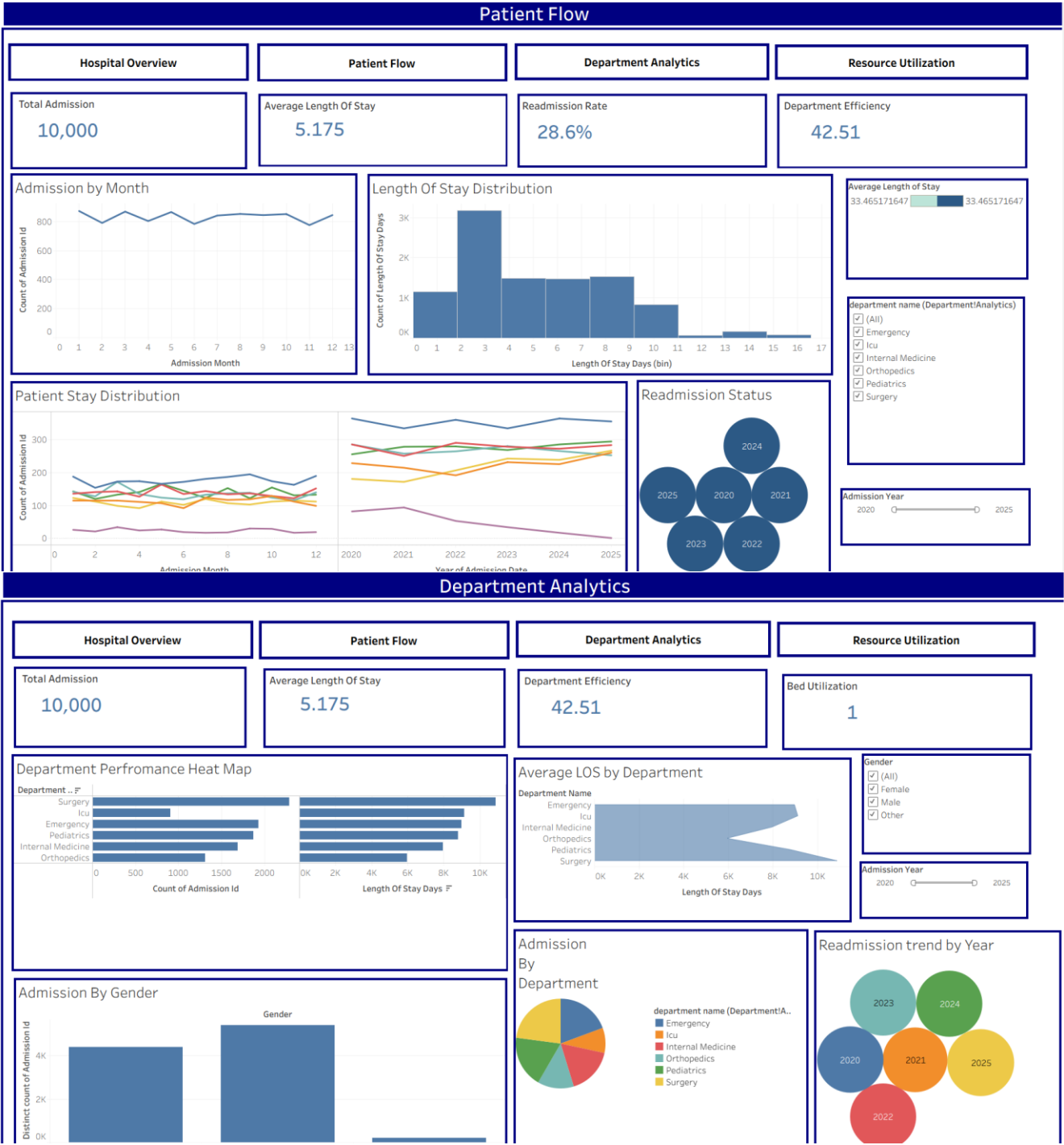
Module 7 contains the final integrated dashboard files and the final validation work.

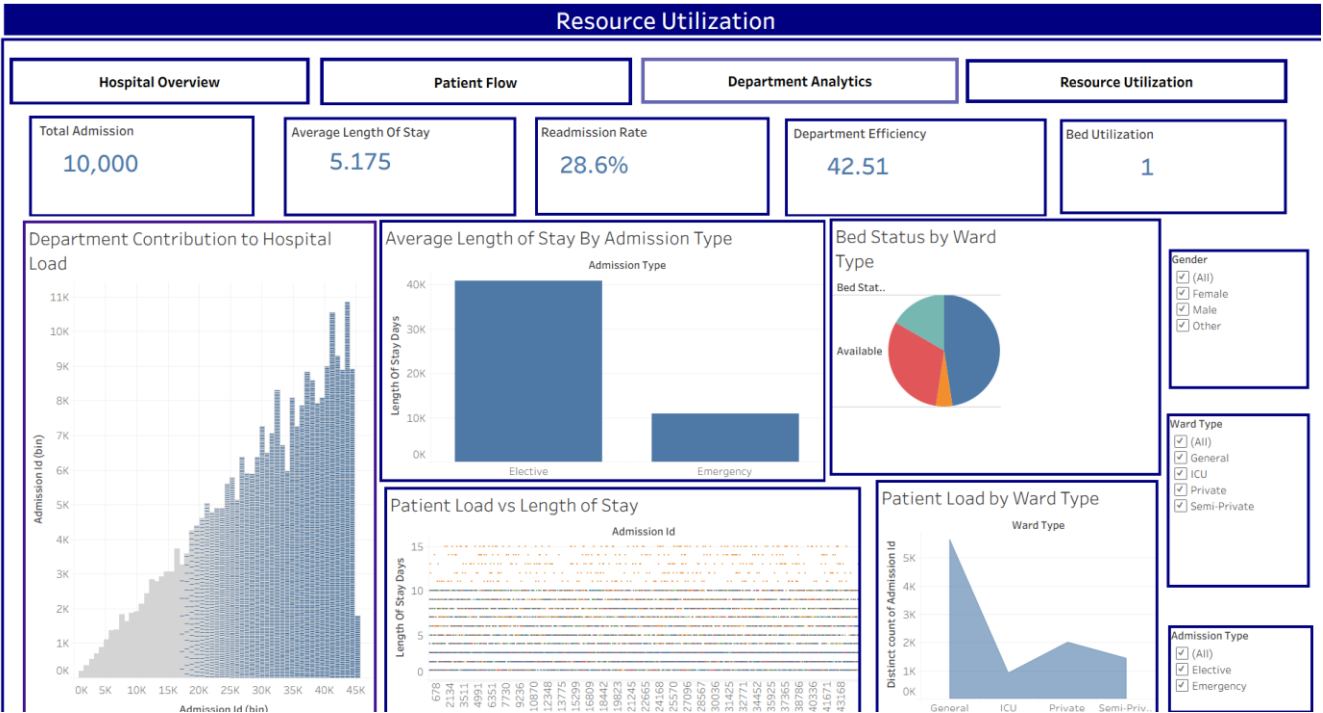
Final file	Purpose
MedTrack_DV.pbix	Final Power BI report.
MedTrack_DV.twbx	Final Tableau workbook containing the dashboard solution.

Testing checklist

Test area	Check
KPI calculations	Values agree with the prepared analytical dataset.
Data consistency	Correct dataset and fields are used.
Filters	Filters affect the intended visuals and KPIs.
Navigation	Navigation works where implemented.
Dashboard actions	Interactive selections behave correctly.
Power BI	Final PBIX opens and visuals/interactions work.
Tableau	Final TWBX opens and dashboards/interactions work.
Final review	Labels, layout, readability and usability are checked.







10. Module 8 – Documentation & Project Delivery

Module 8 organizes the project for final submission and GitHub delivery. Documentation should make it easy to understand the work without opening every project file.

Project identification

Item	Details
Name	Pricilla G
College	Arunachala College of Engineering for Women
Course	B.Tech – Artificial Intelligence & Data Science
GitHub Branch	Pricilla-G
Project	MedTrack_DV – Hospital Operations & Patient Analytics Dashboard

Recommended Module 8 contents

- README.md
- Final project documentation
- Milestone documentation
- Final submission notes, if required by the mentor

11. Recommended GitHub Folder Structure

Based on the files shown in the repository screenshot, the following arrangement is cleaner. It avoids keeping all files in the root and avoids duplicating final dashboard files.

Folder	Files
MODULE 1	data_collection.py hospital_raw_data.csv
MODULE 2	hospital_cleaning.ipynb hospital_cleaned.csv
MODULE 3	generate_hospital_kpis.py hospital_final_dataset.xlsx
MODULE 4	dashboard_Storyboard.pdf
MODULE 5	medtrack_dashboard_v1.pbix
MODULE 6	Module 6 dashboard files/screenshots, if available
MODULE 7	MedTrack_DV.pbix MedTrack_DV.twbx
MODULE 8	Final documentation Milestone documentation
ROOT	README.md

Important corrections before pushing

- Use the same naming style for files and folders. For example, use MODULE 1, MODULE 2, etc.
- Keep the final PBIX and final TWBX in MODULE 7 rather than leaving them mixed with Module 5 files.
- Keep hospital_raw_data.csv in MODULE 1 and hospital_cleaned.csv in MODULE 2.
- Keep the KPI Python script and final Excel workbook together in MODULE 3.
- Keep dashboard_Storyboard.pdf in MODULE 4.
- Do not create fake Module 6 prototype files just to fill the folder. If you have no separate Module 6 file, keep the folder and add the actual screenshots or other Module 6 evidence you used.
- Keep README.md at the repository root.
- Keep the milestone text file in MODULE 8 if it is documentation; rename it from .md.txt to .md only if its contents are actually valid Markdown.

12. Results and Conclusion

Project results

- The cleaned dataset contains 10,000 hospital records.
- Six hospital KPIs were generated.
- The final analytical workbook contains KPI and department/monthly analysis sheets.
- Four dashboard areas were developed.
- Power BI and Tableau files are included in the project.
- The project follows a complete workflow from data preparation to dashboard delivery.

Selected observations from the submitted data

- Surgery has the highest number of admissions, with 2,291 admissions.
- ICU has the highest average length of stay, at approximately 10.08 days.
- Orthopedics has the highest department readmission rate in the submitted department summary, at approximately 30.05%.
- The highest monthly admission count in the submitted monthly data is 162 in March 2025.
- The highest monthly readmission rate in the submitted monthly data is 36.36% in March 2023.

Note

The final KPI workbook reports Occupancy Rate and Bed Utilization Rate as 0.00%. These values are retained as reported in the submitted workbook.

Conclusion

MedTrack_DV demonstrates how hospital data can be prepared, transformed into meaningful KPIs and presented through interactive dashboards. The project combines Python-based data processing with Power BI and Tableau to provide a clear view of hospital admissions, patient flow, department performance and resource utilization.